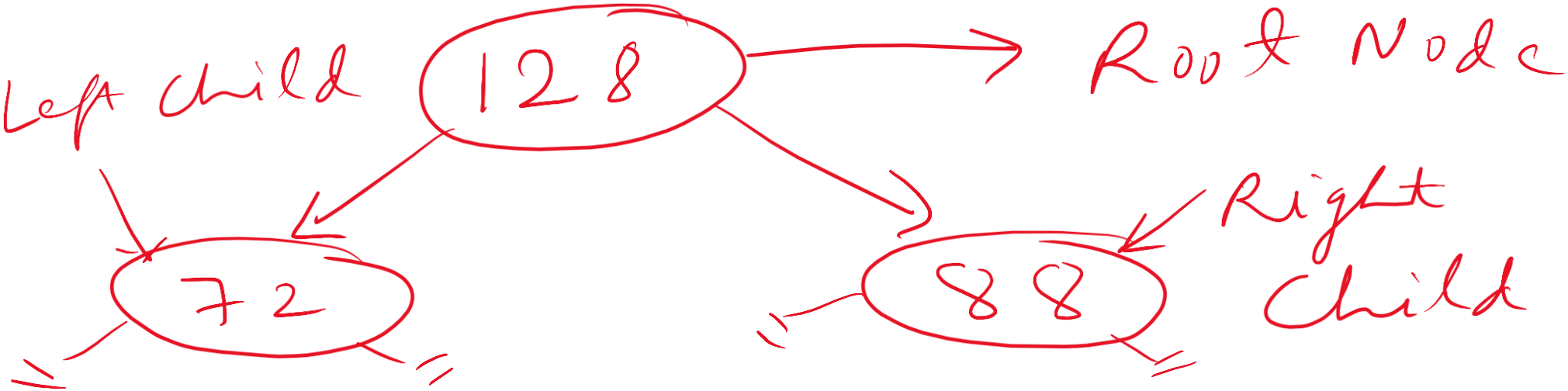
Binary Tree :- It is a data structure in which each Node has at most 2 children i.e left child and right child.

eg :-

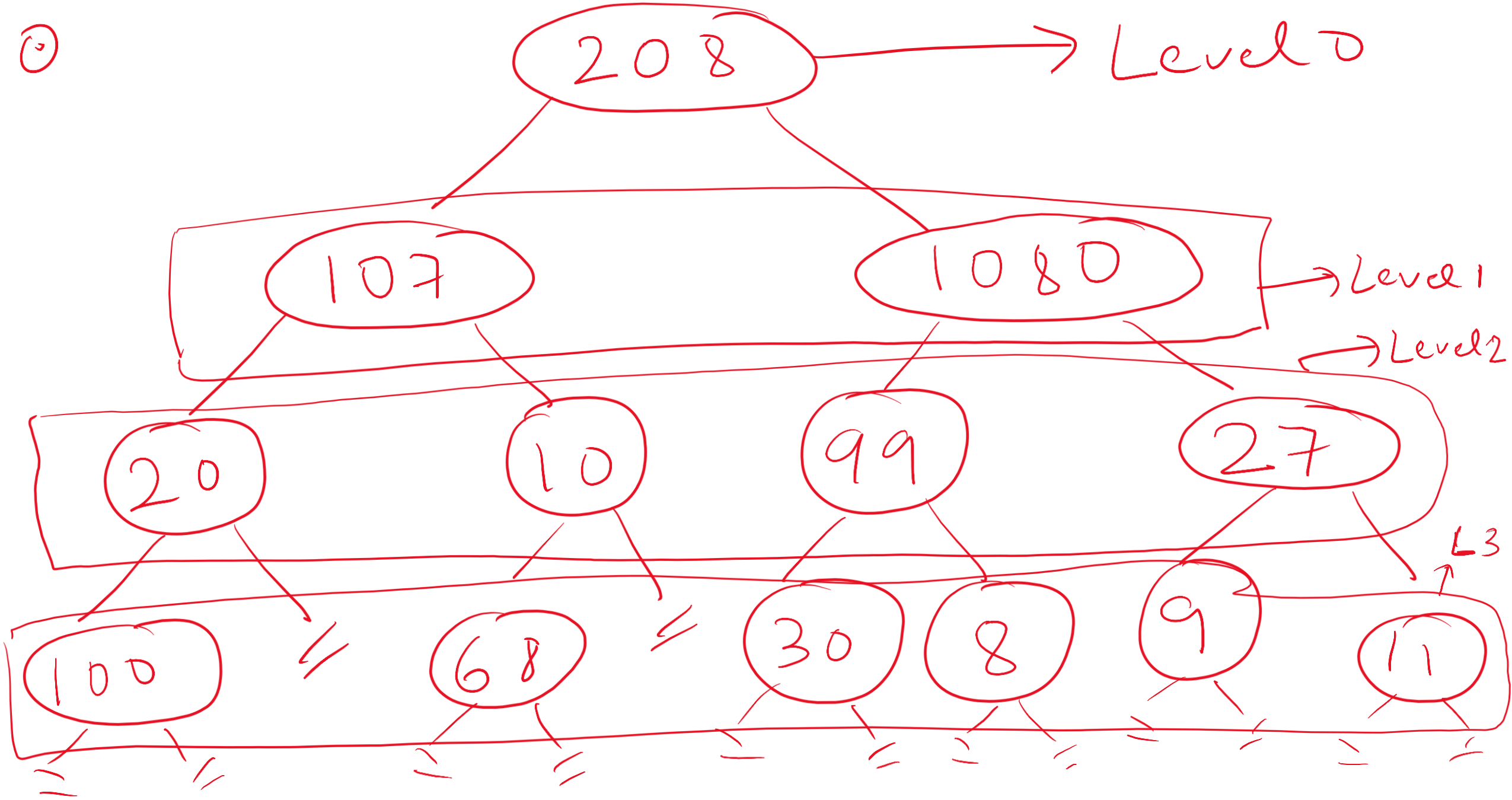


⊙ In Binary Tree, Each Node consists of 3 part :-

1. Data Part
2. Left Child Pointer
3. Right Child Pointer.

- **Root Node** :- It is the topmost Node in a Binary Tree.

- **Leaf Node** :- It is the Node with 0 children.


208
Level 0
107
1080
Level 1
Level 2
20
10
99
27
L3
100
68
30
8
9
11

- All the Questions related to Binary Tree in an Interview can be solved using 2 Techniques:

1. BFS (Breadth First Search)

2. DFS (Depth First Search)

BFS or Level Order traversal :-

Queue (FIFO) :- First In First out

10, 8, 16, 68, 79, 20, 5

Queue :-

~~10~~	~~8~~	~~16~~	~~68~~	~~79~~	~~20~~	~~5~~

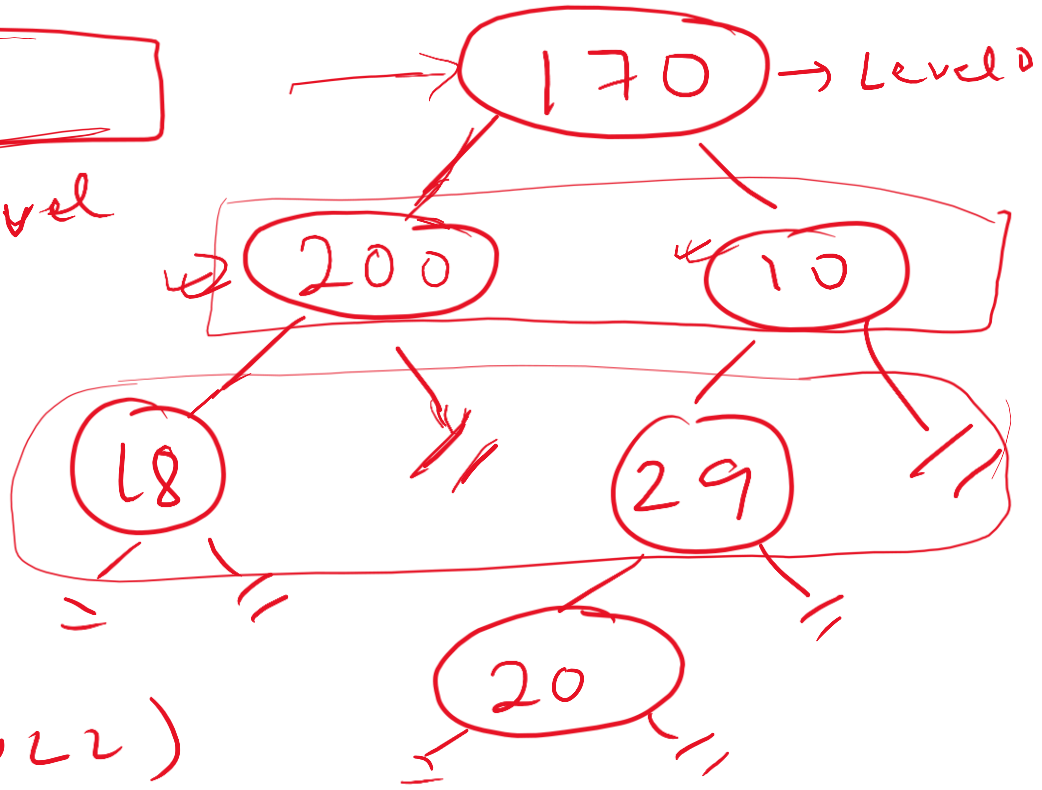
```
Queue<TreeNode> q = new LinkedList<T.N>();
int level=0;
q.add(root);
```

$\boxed{q = \{\ \}}$

```
while(!q.empty())
{
    int size = q.size();
    for(int i=0; i<size; i++)
    {
        T.N front = q.poll();
        if(front.left != NULL)
            q.add(front.left);
        if(front.right != NULL)
            q.add(front.right);
    }//for loop
    level++;
}//w.hile
```

Total no of node at curr level

AB



170, 200, 10, 18, 29, 20

AB → Inside For Loop, It is operating on Nodes that is present at current level.

① " $i = 0$ ⇒ First element of that level

" $i = size - 1$ ⇒ Last element of that level.